# MICROPROCESSOR-BASED INTRUSION DETECTION AND AUTOMATED DOOR CONTROL SYSTEM WITH LCD, TIMERS, LEDs, AND SWITCHES

## Introduction

With the advancement of technology, security systems have evolved from manual systems to automated electronic systems. Traditional security methods such as guards and mechanical locks are inefficient and prone to human error. Microprocessor-based systems provide automated monitoring, faster response, and higher reliability.

Intrusion detection systems are designed to detect unauthorized access and respond appropriately. These systems are widely used in homes, offices, banks, and laboratories. Microprocessors provide intelligence and control by processing signals from sensors and activating output devices such as alarms, displays, and automated doors.

This project focuses on designing a microprocessor-based intrusion detection system with automated door control, LCD display, LED indicators, switches, and timers.

## Problem Statement

Traditional security systems lack automation, real-time monitoring, and reliable response mechanisms. Manual security systems depend on human presence and cannot provide immediate response to intrusion.

There is a need for an automated, reliable, and efficient security system that can:

- Detect intrusion automatically
- Provide visual alerts
- Automatically lock doors
- Display system status
- Operate with minimal human intervention

## Objectives of the Study

### *General Objective*

To design and implement a microprocessor-based intrusion detection and automated door control system.

***Specific Objectives***

- To detect intrusion using motion sensors
- To control door operation automatically
- To display system status using an LCD
- To use LEDs as system indicators
- To implement timers for delay and control
- To provide user control using switches

**Significance of the Study**

- Improving security in buildings
- Reducing human intervention
- Providing automatic response
- Displaying real-time system status
- Enhancing safety and reliability
- Providing a low-cost security solution

**Scope of the Study**

This project focuses on designing and implementing a microprocessor-based intrusion detection system using:

- Motion sensor
- Microprocessor or microcontroller
- LCD display
- LEDs
- Switches
- Automated door mechanism
- Timer functions

The system will detect intrusion and automatically control door locking and status display.

**Parts of the system**

***Microprocessor-Based Systems***

A microprocessor is the central processing unit of a computer system.
It processes input signals and controls output devices based on programmed instructions.
Microprocessors are used in embedded systems such as:

- Security systems
- Automated doors
- Alarm systems
- Smart homes

Advantages include:

- Fast processing
- Automation
- Reliability
- Flexibility

***Motion Sensors***

Motion sensors detect movement within a specific area. The PIR (Passive Infrared) sensor detects infrared radiation from human bodies.
Applications include:

- Security systems
- Automatic lighting
- Automatic doors

***LCD Display in Embedded Systems***

LCD displays are used to display information to users.
Advantages:

- Low power consumption
- Easy to interface

- Clear display

***Automated Door Control Systems***

Automated doors use motors or solenoids to open and close automatically. These systems improve security and convenience.

***Timer in Microprocessor Systems***

Timers are used to control delays and timing operations. They allow systems to perform operations after specific time intervals.
Applications include:

- Delay control
- Alarm duration
- Automatic door closing

**Memory in the system**

The system as tow types of memories; ROM memory and R/W memory

***ROM (Program Memory)***

Functions:

- Stores permanent program
- Stores system logic
- Provides instructions to CPU

***RAM (Data Memory)***

Functions:

- Stores temporary data
- Stores sensor status
- Stores timer values

### *System Bus*

The **system bus** is a communication pathway that connects the **microprocessor**, **RAM**, **ROM**, and all input/output devices (LCD, sensors, LEDs, door motor, switches, and timer). It allows data and control signals to move between these components.

## System Methodology

### *System Overview*

The system consists of input devices, processing unit, and output devices.

*Inputs:*

- PIR sensor
- Switches

*Processing unit:*

- Microprocessor
- R/W memory
- ROM memory

*Outputs:*

- LCD display
- LEDs
- Automated door

*Timer:*

- Controls delay and timing operations

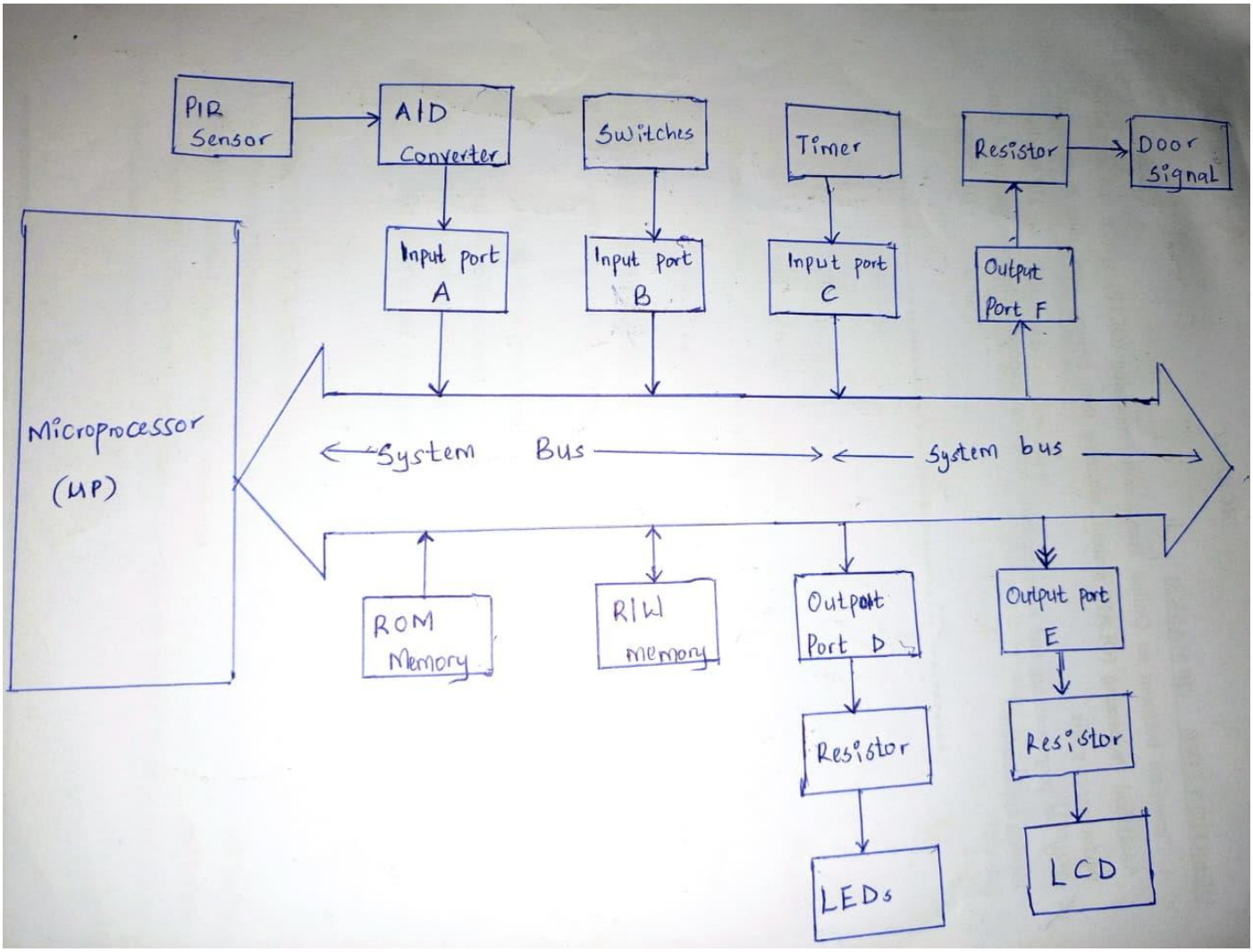
*A/D converter*

- Used for converting the analog signals from the PIR sensors to digital signals to send into the microprocessor.

*Resistors*

- Used for pulling up the electrical power from the system to activate the actuators in the door signal, the LEDs and the LCD

**System Block Diagram**




**System Operation**

*Normal condition:*

- Green LED ON
- LCD displays system ready
- Door unlocked

*Intrusion detected:*

- Red LED ON
- LCD displays intrusion alert
- Door locked automatically

- Timer activated

*Authorized entry:*

- Switch pressed
- Door opens
- Timer closes door automatically

**Software Design**

The system uses programmed instructions to control operation.

Steps:

- Read sensor input
- Process input against the instructions from the ROM memory.
- Control outputs
- Store the status into the R/W memory.
- Display status
- Control timer

**Conclusion**

The microprocessor-based intrusion detection system provides an effective and reliable security solution. The system can detect intrusion, display system status, and automatically control doors. The use of timers improves automation and efficiency.

The system is suitable for homes, offices, and secure facilities.

**Recommendations**

Future improvements may include:

- Adding alarm buzzer
- Adding GSM module for notifications
- Adding password authentication
- Adding IoT for remote monitoring